FUTURE UNIVERSITY

Department of Computer Science & Engineering

PROJECT SYNOPSIS
B. Tech Final Year Project
Academic Year: 2025-26

PROJECT TITLE:

AgroGuard

An Intelligent Deep Learning System for Real-time Crop Disease Detection and Management

TEAM DETAILS:

TEAM MEMBER 1: -

- **NAME:** Vaishnavi Dixit
- **ROLL NUMBER:** 2204760100112
- **EMAIL ID:** vaishnavidixit15@gmail.com
- **CONTACT NO:** 8009679005

TEAM MEMBER 2: -

- **NAME:** NOMAN ALI KHAN
- **ROLL NUMBER:** 2204760100075
- **EMAIL ID:** devnomanali@gmail.com
- **CONTACT NO:** 7505978684

TEAM MEMBER 3: -

- **NAME:** YASHWANT CHAUHAN
- **ROLL NUMBER:** 2204760100118
- **EMAIL ID:** Yashwantc19@gmail.com
- **CONTACT NO:** 7060143862

Project Guide Name: Dr. Sachin Raj Saxena
Project Guide Designation:

ABSTRACT

Agriculture is the backbone of the economy, yet crop diseases remain a significant threat to global food security, often resulting in up to 40% yield loss. Crop diseases significantly reduce agricultural productivity and are responsible for major economic losses worldwide. Early detection is critical but remains challenging due to limited expert availability and slow diagnostic processes. AgroGuard is proposed as a deep learning powered intelligent system that performs real-time crop disease detection using leaf image analysis. The system leverages Convolutional Neural Networks and transfer learning techniques to classify plant diseases with high accuracy across multiple crop types.

The project aims to provide a user-friendly web interface enables farmers and agricultural workers to upload plant images and receive instant diagnostic results along with scientifically grounded treatment and prevention recommendations. The platform integrates a modern frontend with a scalable backend API and a structured database for disease knowledge and user history. The system aims to reduce dependency on manual inspection, improve response time, and encourage optimal pesticide usage. The system ensures low latency and high scalability.

Overall, AgroGuard serves as a practical AI-driven decision support tool for smart agriculture and sustainable crop management

INTRODUCTION

- Background

Agriculture remains one of the most critical sectors supporting human survival and economic development. In many regions, especially developing countries, a

large portion of the population depends directly on farming for livelihood. However, plant diseases continue to be a primary factor causing crop damage, yield reduction, and quality degradation.

- Motivation

With the advancement of Artificial Intelligence and Deep Learning, image-based classification systems have shown strong performance in medical and agricultural diagnostics. Convolutional Neural Networks can automatically learn complex visual patterns and features directly from images, making them suitable for plant disease detection.

The rapid spread of smartphones and rural internet connectivity provides an opportunity to deploy AI-powered diagnostic systems through web platforms. Farmers can capture images using mobile devices and receive expert-level analysis within seconds

- Need for the Proposed System

Traditional disease detection methods rely heavily on visual inspection by trained experts or laboratory-based testing. These methods are not only slow and expensive but are also difficult to access for small and remote farmers. Visual similarity between different diseases further increases the chances of incorrect diagnosis.

- Current scenario/existing systems

Existing commercial solutions are often costly, closed-source, or restricted to limited crops. Many research systems are not production-ready. Therefore, there is a strong need for an open, scalable, accurate, and user-friendly platform. AgroGuard is designed to address these challenges through modular AI architecture, transfer learning models, and an accessible web interface.

PROBLEM STATEMENT:

Farmers often lack immediate access to agricultural experts, leading to delayed or incorrect diagnosis of crop diseases. This leads to the misuse of fertilizers and the spread of infection, causing massive financial losses and environmental degradation. There is a strong requirement for an automated, scalable, and accurate disease detection system that works using simple image inputs and provides actionable treatment guidance.

OBJECTIVES:

- Develop a robust image-based plant disease detection pipeline.
- Collect, clean, and preprocess a large labelled dataset.
- Apply transfer learning using state-of-the-art CNN architectures.
- Optimize model accuracy, precision, and recall metrics.
- Design a responsive and accessible web interface.
- Build a Flask-based prediction API for real-time inference.
- Create a treatment recommendation mapping engine.
- Maintain disease knowledge and user scan history database.
- Ensure scalability and modular deployment.

SCOPE OF THE PROJECT:

What it covers: Supports detection of multiple diseases across selected crops. Uses only leaf image inputs for prediction. Provides instant classification and confidence score. Displays treatment and prevention guidelines. Maintains user diagnosis history.

What it does NOT cover: Soil nutrient analysis or irrigation scheduling. Does not include satellite or drone imagery.

Future Scope: Integration of IoT sensors for real-time soil health monitoring and a mobile app version.

PROPOSED METHODOLOGY:

The system follows a Modular Architecture:

- **Data Acquisition:** The system follows a structured AI development lifecycle starting from dataset acquisition to deployment. Standard public datasets such as Plant Village are used as the primary training source.
- **Preprocessing:** Image resizing, normalization, and data augmentation (rotating/flipping) to improve model generalization.
- **Model Training:** Implementing **Transfer Learning** using a pre-trained **ResNet50** or **InceptionV3** architecture for high accuracy with lower computational cost.
- **Backend Integration:** The trained model is exported and wrapped inside a Flask REST API. The API accepts image input and returns prediction results in JSON format. The React frontend communicates with the API and displays predictions, disease details, and treatment advice.
- **Database**: A MongoDB database stores disease metadata, recommendation text, and user scan history.

TECHNOLOGY STACK:

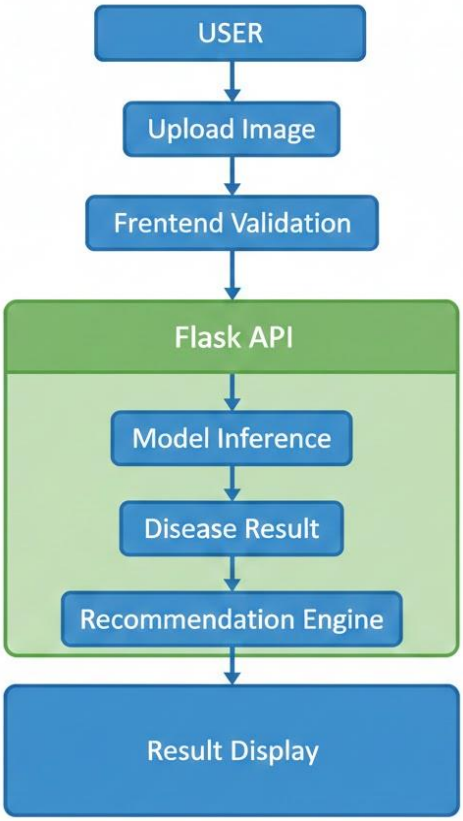
- **Frontend:** React.js, Tailwind CSS, Axios
- **Backend:** Python Flask REST API
- **Database:** MongoDB (for storing user history and disease data)
- **ML Libraries:** TensorFlow/ Keras, OpenCV, NumPy, Pandas
- **Tools:** Git, Docker, Postman, Jupyter
- **Languages:** Python, JavaScript

MODULES/FEATURES:

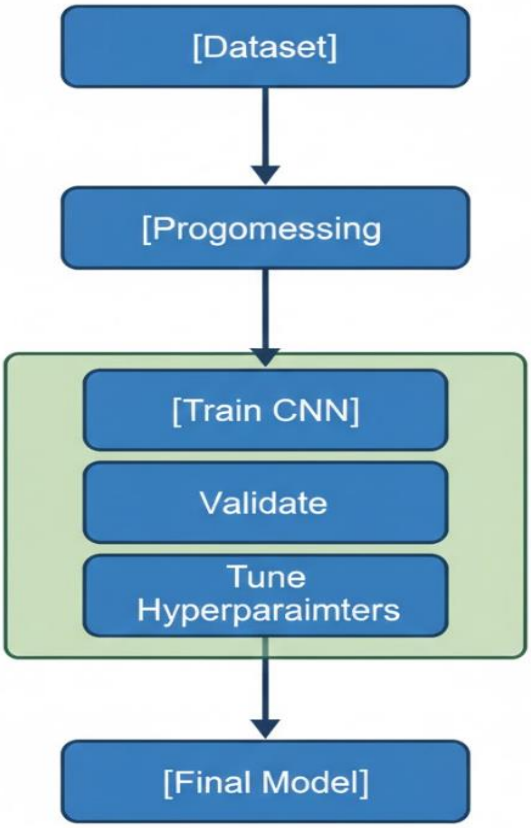
1. **User Authentication:** Secure login for farmers to track their crop history.
2. **Disease Diagnosis:** An upload portal for image analysis using the trained ML model.
3. **Agro-Wiki:** A searchable database of diseases, symptoms, and prevention tips.
4. **Prediction Engine –** Detailed functional module with logging and validation.
5. **Admin Control Panel** – Detailed functional module with logging and validation.

ARCHITECTURAL DIAGRAM:

System Flowchart:



Model Training Flowchart:



PROJECT TIMELINE:

Phase	Activity	Duration	Output
1	Literature Survey	3 Weeks	Problem Understanding
2	Dataset Prep	4 Weeks	Clean Dataset
3	Model Training	4 Weeks	Base Model
4	Model Tuning	3 Weeks	Optimized Model
5	Backend API	3 Weeks	Flask API
6	Frontend UI	3 Weeks	React App
7	Integration	2 Weeks	Working System
8	Testing	2 Weeks	Validated System
9	Documentation	2 Weeks	Final Report

SYSTEM REQUIREMENTS:

Hardware

- Processor: Intel i5 or higher (GPU recommended for training)
- RAM: 8GB (minimum), 16GB (recommended)
- Storage: 256GB SSD
- GPU: Optional but recommended for training
- OS: Windows/Linux
- Server: Flask runtime environment

Software

- OS: Windows 10/11 or Linux (Ubuntu)
- Software: Python 3.x, Node.js, VS Code
- Development Tools: VS Code, Jupyter Notebook, Postman

EXPECTED OUTCOMES:

1. A high-accuracy model (target >92%) for crop disease classification.
2. A functional, real-time web interface for easy accessibility.
3. Reduced turnaround time for disease diagnosis for end-users.
4. Reusable ML deployment pipeline
5. Better pesticide decisions

REFERENCE:

• Plant Village Dataset Paper

• TensorFlow Official Docs

• Keras Transfer Learning Guide

• IEEE Papers on Plant Disease CNN

• FAO Crop Disease Reports

Student Signature(s): __Vaishnavi_, _Noman_, _Yashwant___

Guide Signature: _____